BiFrost

iGEM IIT Bombay

Attributions :

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Sponsors

WADHWANI | IIT BOMBAY Innovation Centre Health and Bio

Engineering Yeast to produce Bio-Coolant for Sustainable Data Centre Cooling

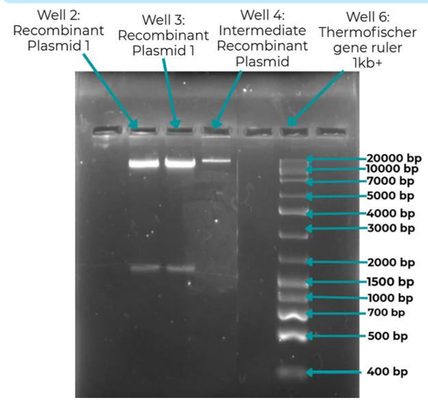
PROBLEM STATEMENT

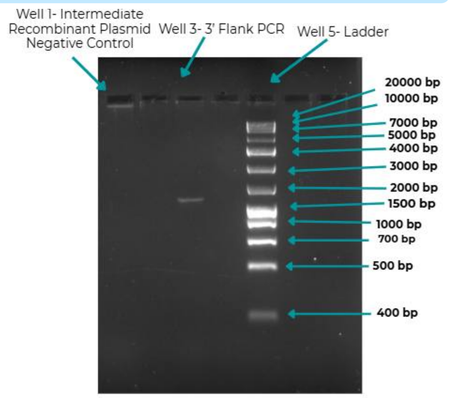
- The exponential growth of AI and cloud computing has made data centres one of the **fastest-growing consumers** of electricity. Approximately 40% of this energy is spent not on computation, but on cooling servers to maintain reliable performance.
- **PFAS-based dielectric fluids** set the benchmark for immersion cooling because of their outstanding thermal and electrical performance. However, their **environmental persistence** has driven the industry toward petroleum-derived mineral oils, which remain **fossil-based** and often compromise on performance or sustainability. This creates a need for a renewable coolant with the efficiency of PFAS and the sustainability of bio-based materials.

OUR SOLUTION

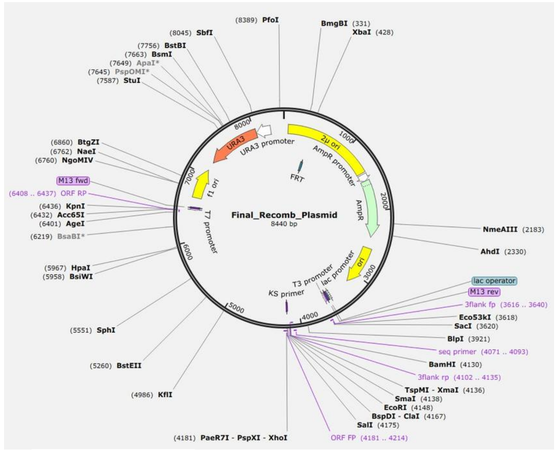
- We identified **methyl myristate** as a promising **bio-based coolant** candidate because of its favorable thermophysical & dielectric properties.
- Instead of conventional petrochemical synthesis, we engineer ***Saccharomyces cerevisiae*** to biosynthesise methyl myristate through fermentation by rewiring its native lipid metabolism toward chain-length-specific C14 ester production.
- *Our microbial platform enables **sustainable, scalable, and cost-effective production** of methyl myristate via fermentation.*
- *Combined with performance-enhancing additives, it provides a **renewable alternative** to conventional dielectric coolants for next-generation green data centres.*

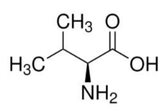
Check for final recombinant plasmid





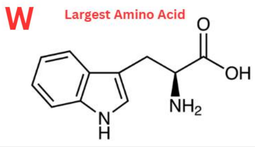
PLASMID MAP





V 1254 W

Largest Amino Acid



PFAS: The "Forever Chemicals"

- PFAS (Per- and Polyfluoroalkyl Substances) are **synthetic fluorinated compounds** widely used in immersion cooling for their excellent dielectric strength, thermal stability, and chemical inertness.
- However, their strong carbon–fluorine bonds make them highly persistent in the environment, leading to **bioaccumulation** and increasing global regulatory restrictions.

From Italy to India: A Cautionary Tale

Following the closure of production in **Italy** after PFAS contamination of one of Europe's largest aquifers, its fluorochemical equipment was **relocated to Ratnagiri, India.** The incident affected **350,000+ people** through contaminated drinking water and has reignited concerns about shifting hazardous industries to regions with less stringent regulations.

THE PFAS PARADOX

For decades, **PFAS-based fluids** set the benchmark for immersion cooling due to their excellent dielectric strength, thermal stability, and non-flammability. However, the same carbon–fluorine bonds that make them industrially valuable also make them among the most persistent synthetic chemicals ever produced.

Concerns

Environmental Persistence

- PFAS are often called **"forever chemicals"** because they resist natural degradation and accumulate in soil, groundwater, and living organisms.
- Scientists have suggested that PFAS contamination has contributed to **exceeding planetary chemical pollution boundaries**.

Human Health

- Long-term exposure has been associated with kidney and testicular **cancers**, liver and thyroid **dysfunction**, reproductive **disorders**, and immune suppression.

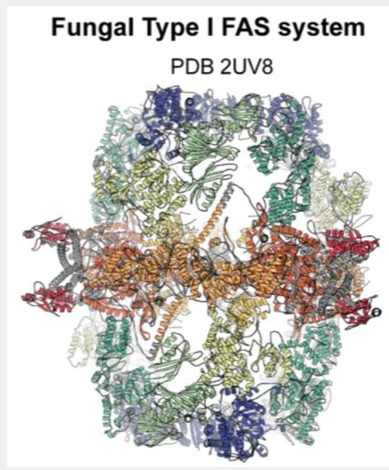
An Industry in Transition

- In 2022, 3M announced the complete phase-out of PFAS manufacturing, including the widely used Largest PFAS producing factory product lines, creating one of the **largest supply gaps in immersion cooling history.**

Global Regulatory Shift

- The EU is introducing **strict PFAS limits** and evaluating restrictions covering 10,000+ PFAS compounds.
- The **U.S. EPA** has established enforceable drinking water standards, accelerating the global transition away from PFAS.

Methyl myristate meets the majority of **OCP immersion coolant requirements**, with **additive engineering** enabling compliance with the remaining performance criteria.



WHAT IS FAS 2?

In yeast, FAS stands for **Fatty Acid Synthase.** It is a multi part molecular machine/ enzyme complex whose sole job is to build fat molecules (lipids) inside the cell.
FAS2 gene **encodes the alpha subunit** of the FAS complex. This complex is the **primary biological machine** responsible for the biosynthesis of long chain fatty acids.

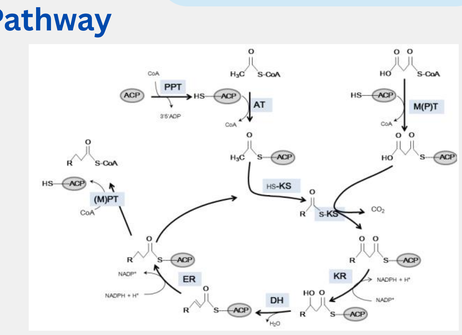
Why Target C-14 FFA?

Standard biological fatty acids (C16-C18) do not always have the **optimal physical properties** for industrial applications. By shortening the carbon chain length precisely to C14, we alter the **melting point and kinematic viscosity** of the resulting lipids.

FFA Synthesis Pathway

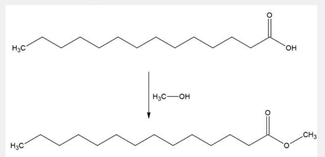
Crucially, the alpha subunit houses the KS domain, which serves as the **core catalytic site** regulating carbon chain elongation.

In wild type yeast, the KS domain natively **accommodates the growing acyl chain** until it becomes C16-C18.



Property	Methyl Myristate
Viscosity	~3.5–3.9 cSt
Boiling Point	~295–323°C
Specific Heat	~1.9–2.1 J/g·K
Dielectric Constant	~3.1 - 3.3
Melting Point	~18 - 19°C
Flash Point	~157-160°C

Methyl Myristate



Experiment Design-1

- **Gene Extraction (S. cerevisiae):** We source our wild-type target gene directly from our chosen chassis, **Saccharomyces cerevisiae BY4742**. The specific sequence we extract is the FAS2 gene.
- **Mutation Place:** The precise target is the **KS** domain within **FAS2.** We engineer a **V1254W** substitution. Steric hindrance induced by Tryptophan halts chain elongation at **C14**
- **Plasmid Vector:** We use the **high-copy p426GPD** vector backbone **(~5.8 kb)**. This plasmid acts as our primary vehicle to eventually deliver the mutated FAS2 gene back into the yeast.

Experiment Design-2

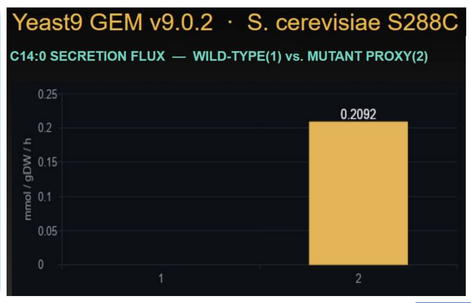
- **Inserts & Assembly:** We sequentially ligate the engineered FAS2 **ORF (~5.6 kb) (KpnI/XhoI)** and the **3' Flank (~0.5 kb) (BamH1/SacI)** into the vector. This assembly replaces the vector's native **GAP promoter** with our custom inserts,.
- **Bacterial Amplification:** The assembled construct is first transformed into an intermediate host, **E. coli DH5α,** to biologically amplify the plasmid for testing & yeast transformation.
- **Checks:**
  - **Colony PCR** visually confirms the integration of the **~0.5 kb 3' Flank.**
  - A **HindIII** restriction digest confirms the presence and correct orientation of the **~5.6 kb FAS2 ORF** via diagnostic gel images.
  - Finally, the plasmid undergoes full Sequencing to guarantee **100% fidelity of the V1254W mutation site.**

Experiment Design-3

- **Extraction from Bacteria:** Once the recombinant plasmid (now totaling **~11.9 kb including the vector, ORF, and flank**) is fully validated, we perform a Miniprep to extract & purify the complete, amplified plasmid from the E. coli cloning host.
- **Transformation Protocol:** We take this purified plasmid and transform it directly into our primary production chassis: Saccharomyces cerevisiae BY4742.
- **Selection & Output:** The transformed yeast are plated on selective media (utilizing the **URA3 selection marker** inherent to the **p426GPD plasmid**). The surviving colonies are cultivated, harvested, and lysed so their lipid extracts can be analyzed via **LC-MS** to definitively quantify the successful overproduction of **C14 Myristic Acid.**

FLUX BALANCE ANALYSIS

- Lipid bio-synthesis in yeasts is a highly regulated metabolic network, favoring C16/C18 fatty acids (FA) production. We require a very specific lipid required for our target applications- **Myristic acid (C14:0 FA)** but wild-type **S. cerevisiae** produces very trace amounts (~1%) of C14 FA.
- We built a constraint-based computational model using the **Yeast8 Genome-Scale Metabolic Model** (GEM v8.0.2). By adding a **C14:0 exchange reaction** and enforcing a >10% viability constraint, we can **explicitly simulate** and **predict** how genetic changes affect the organism's FA secretion profile.
- Using linear algebra techniques with constrained optimization, a GLPK solver maximizes our C14:0 flux output. The model reveals that forcing C14:0 export requires **reversing acyl-CoA ligase activity** and **exploiting C16:0 degradation**, actively pulling metabolic flux away from standard lipid sequestration.



SEQUENCE ALIGNMENT

- **BLAST of the ScFAS2** KS domain against **fungal FAS** sequences returns Y. lipolytica as the top hit by sequence identity, the closest characterized homolog with validated tunnel residues.
- Alignment maps the **three YlFAS tunnel residues** onto **ScFAS2 (M1217 to M1251, I1220 to V1254, M1226 to M1260)**, done with steric hindrance in mind.
- **M1251 and M1260** are **conserved in 105/109 KS-family** enzymes, so we left them alone. V1254 (I1220 in Yl) is poorly conserved & differs between **species (Val in Sc, Ile in Yl)**, making it our actual engineering target.



Immersion Cooling

Immersion cooling is a cooling technology in which electronic components such as CPUs, GPUs, and servers, are fully submerged in a specially engineered, electrically non-conductive (dielectric) liquid instead of being cooled by air.

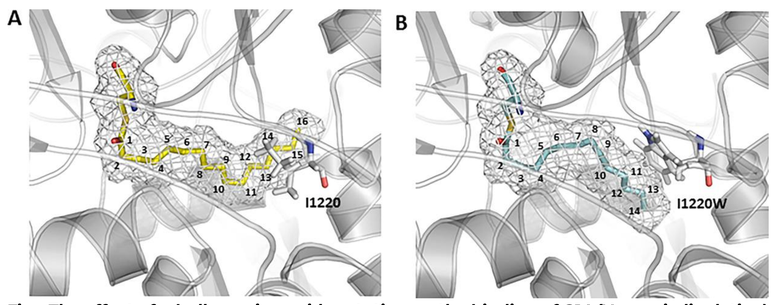
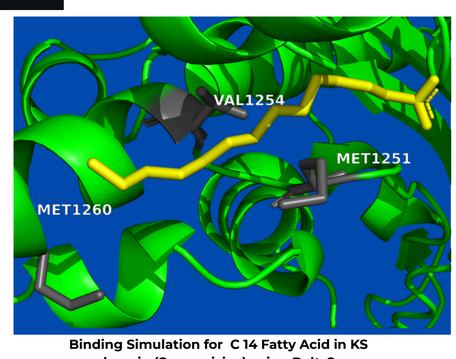


Fig - The effect of a bulky amino acid mutation on the binding of C14 (Yarrowia lipolytica)
https://pubs.acs.org/asbcd6/article/6/10/1870/1264995/Production-of-Medium-Chain-Fatty-Acids-by-Yarrowia.



Binding Simulation for C 14 Fatty Acid in KS domain (S. cerevisiae) using Boltz2.

Position	Residue	Conservation Score
1251	M	0.565
1254	V	0.372
1260	M	0.764

Conservation Score for KS Domain of *Saccharomyces cerevisiae*

THE HIDDEN BOTTLENECK: TODAY'S IMMERSION COOLANTS

Current immersion cooling fluids are built on fragile petrochemical supply chains

42% of the single-phase immersion cooling market relies on synthetic hydrocarbon fluids (PAO/GTL), yet these fluids originate from manufacturing pathways developed for lubricants and fuels, not immersion cooling.

Why is this a concern?

Highly concentrated supply chain

PAO

- Just three manufacturers control ~91% of global PAO production capacity, **limiting supplier diversity** and increasing the **risk of shortages and price volatility.**

GTL

- Industrial-scale GTL technology is licensed by only two companies worldwide, while only five major GTL plants supply the global market, making **expansion slow** and **vulnerable to geopolitical events.**

Climate-sensitive manufacturing

- >95% of U.S. ethylene production which is the key feedstock for PAO, is **concentrated** along the Gulf Coast.
- **Multiple events,** including Hurricane Harvey (2017), Hurricane Laura (2020), the Texas Freeze (2021), and Winter Storm Elliott (2022), have **repeatedly disrupted production.**

AI is accelerating demand

- This structural weakness will translate into coolant shortages, longer lead times, and higher costs as immersion cooling moves up from a **$178 million** market to a multibillion dollar market.

Integrated Human Practices

Consulted Over 30+ Stakeholders

Academia Connect
1. Metabolic Aspect
2. Coolant Perspective
Consulted: Prof. Balaji, Prof. Sandeep, Prof. Phale, Prof. Suyash, Diagnobacs (Ms. Rutuja), Pratik Makhija (IISER Pune), Alums (Kadambari, Husayn, Deepan, and Dawar)
Takeaway: Refined our metabolic engineering strategy, pathway design, and experimental approach to improve the robustness of our project.

Industry Connect
Consulted: Shell, BioCompute, ICT, Dr. Rajesh Vagdama(L'Oréal), Vaibhav Sai and WITC
Takeaway: Identified key industry expectations regarding coolant performance, scalability, manufacturability, and real-world implementation.

Global Collaboration
Consulted: iGEM Denmark and Nicola O'Riordan.
Takeaway: Incorporated international perspectives, best practices, and collaborative insights to further refine our project design and implementation.



Educational Outreach

SynBioCon 2026 (IITB X IITKGP)
International virtual conference featuring global experts and a hands-on bioinformatics workshop.

Gnaan U 2026
Inspired young minds to explore biology through engaging demonstrations, discussions, and experiential learning activities





Within Campus (Enthuse, orientation etc)
Showcased our project through campus wide events, engaging students in discussions on synthetic biology, sustainability, and research.

TechConnect
Presented our research through an interactive Techfest exhibit, engaging visitors in conversations on synthetic biology and its real-world applications.



Additional Initiatives: AI Audit • Interactive Gene Cloning Simulator • International Level Public Awareness Survey

Team BiFrost — Aarohi Gupta, Angel Singhvi, Ranit S sooraj, Madhur Likhmani, Durva Saraf, Arhan Khade, Yashika Singh, Sagnik Dey, Arnav Oza, Bipra Bhanu Mohanty, Shinjini Jain, Tanish Jain, Latisha Meena, Abhinav Gupta, Hardika Jain, Aayush Gupta, Manish S, Ishaan Chhaaya, Siddhant Chowdhary, Gururatna Upadhyay, Mitali Arya, Pratik Ingle, Nisarg Rathod, Aryan Prasad